JOINS AND SET OPERATIONS

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Create a database & use:

```
1   CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 • USE bookflow_db;
```

Output

Action Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db

Create a table

```
3  • CREATE TABLE class (
4        id INT,
5        name VARCHAR(30)
6    );
7  • CREATE TABLE class_info (
8        id INT,
9        address VARCHAR(30)
10   );
```

Output

Action Output

#	Time	Action
1	09:47:38	CREATE DATABASE IF NOT EXISTS bookflow3_db
2	09:47:55	USE bookflow3_db
3	09:48:21	CREATE TABLE class (id INT, name VARCHAR(30))
4	09:48:23	CREATE TABLE class_info (id INT, address VARCHAR(30))

Insert data

```
12 • INSERT INTO class VALUES
13   (1,'Likhitha'),
14   (2,'Tejaswini'),
15   (4,'Ganesh');
16 • INSERT INTO class_info VALUES
17   (1,'KERALA'),
18   (2,'ANDRA PRADESH'),
19   (3,'CHENNAI');
```

7	12:17:43	INSERT INTO class VALUES (1,'Likhitha'), (2,'Tejaswini'), (4,'Ganesh')
8	12:17:45	INSERT INTO class_info VALUES (1,'KERALA'), (2,'ANDRA PRADESH'), (3,'CHENNAI')

Show data

```
20 •  SELECT * FROM class;
```

1	Likhitha
2	Tejaswini
4	Ganesh

```
21 •  SELECT * FROM class_info;
22
```

1	KERALA
2	ANDRA PRADESH
3	CHENNAI

Cross join

```
22 •  SELECT *
23    FROM class
24    CROSS JOIN class_info;
25
```

1	Likhitha	3	CHENNAI
1	Likhitha	2	ANDRA PRADESH
1	Likhitha	1	KERALA
1	Likhitha	3	CHENNAI
1	Likhitha	2	MUMBAI
1	Likhitha	1	DELHI
2	Tejaswini	3	CHENNAI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	1	KERALA
2	Tejaswini	3	CHENNAI
2	Tejaswini	2	MUMBAI
2	Tejaswini	1	DELHI
4	Ganesh	3	CHENNAI
4	Ganesh	2	ANDRA PRADESH
4	Ganesh	1	KERALA
4	Ganesh	3	CHENNAI
4	Ganesh	2	MUMBAI

Inner join

```
25 •  SELECT *
26    FROM class
27    INNER JOIN class_info
28    ON class.id = class_info.id;
```

1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI
1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI

```
29 •  SELECT class.name,
30            class_info.address
31    FROM class
32    INNER JOIN class_info
33    ON class.id=class_info.id;
```

Likhitha	KERALA
Likhitha	DELHI
Tejaswini	ANDRA PRADESH
Tejaswini	MUMBAI
Likhitha	KERALA
Likhitha	DELHI
Tejaswini	ANDRA PRADESH
Tejaswini	MUMBAI

Natural join

```
34 •  SELECT *
35    FROM class
36    NATURAL JOIN class_info;
```

1	Likhitha	KERALA
1	Likhitha	DELHI
2	Tejaswini	ANDRA PRADESH
2	Tejaswini	MUMBAI
1	Likhitha	KERALA
1	Likhitha	DELHI
2	Tejaswini	ANDRA PRADESH
2	Tejaswini	MUMBAI

Left Outer join

```
37 •  INSERT INTO class VALUES
38    (5,'Shiva');
39 •  INSERT INTO class_info VALUES
40    (7,'NOIDA'),
41    (8,'PANIPAT');
42
```

Output

Action Output

#	Time	Action
18	12:27:54	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')
19	12:27:58	INSERT INTO class VALUES (5,'Shiva')
20	12:28:00	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')

Left join

1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI
4	Ganesh	NULL	NULL
5	Shiva	NULL	NULL

```sql
SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;
```

Right join

```sql
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;
```

NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Union

```sql
CREATE TABLE first_table(
id INT,
name VARCHAR(30)
);
CREATE TABLE second_table(
id INT,
name VARCHAR(30)
);
INSERT INTO first_table VALUES
(1,'Sita'),
(2,'Sai');
INSERT INTO second_table VALUES
(2,'Sai'),
(3,'Ram');
```

30	12:34:31	INSERT INTO first_table VALUES (1,'Sita'), (2,'Sai')
31	12:34:33	INSERT INTO second_table VALUES (2,'Sai'), (3,'Ram')

Union

```sql
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

Result Grid

id	name
1	Sita
2	Sai
3	Ram

Union All

```
72 •  SELECT * FROM first_table
73    UNION ALL
74    SELECT * FROM second_table;
75
```

Result Grid | Filter Rows: | Expor

id	name
1	Sita
2	Sai
2	Sai
3	Ram

Intersect

```
76    INTERSECT
77    SELECT * FROM second_table;
78
```

Result Grid | Filter Rows:

id	name
2	Sai

Minus

```
78 •  SELECT first_table.*
79    FROM first_table
80    LEFT JOIN  second_table
81    ON  first_table.id=second_table.id
82    WHERE second_table.id IS NULL;
83
```

Result Grid | Filter Rows: | Expo

id	name
1	Sita

Address availability status:

```
82 •  SELECT c.id,
83          c.name,
84          CASE
85              WHEN ci.address IS NULL
86              THEN 'Address Missing'
87              ELSE 'Address Available'
88          END AS Status
89    FROM class c
90    LEFT JOIN class_info ci
91    ON c.id=ci.id;
92
```

Result Grid | Filter Rows: | Export:

id	name	Status
1	Likhitha	Address Available
1	Likhitha	Address Available
2	Tejaswini	Address Available
2	Tejaswini	Address Available
4	Ganesh	Address Missing